

Assignment 6 – B – Ring Algorithm

AIM: Implement Ring Algorithm for leader election.

Objective: To elect a coordinator process in a distributed system using ring election algorithm.

Outcome: Current active process elects the process with highest priority as coordinator using ring structure.

Explanation: The ring algorithm is a type of election algorithm used in distributed systems for selecting a coordinator process among active processes arranged in a logical ring.

In a distributed system, a coordinator is required to manage operations and perform functions needed by all other processes.

Election algorithms are used to select one process as coordinator when the existing coordinator crashes or becomes unavailable.

In ring algorithm, all processes are arranged in a logical ring, and each process knows the identity of its next process in the ring.

Every process has a unique priority number or process ID.

The process having the highest priority number among active processes is selected as coordinator.

When a process detects that the coordinator has failed, it starts an election process.

The Ring algorithm works as follows:

Let us assume that process P detects that the current coordinator is not responding.

Process P creates an election message containing its own process ID.

This election message is passed to the next active process in the ring.

Each active process receives the message, adds its own process ID to the election message, and forwards it to the next process.

The message continues circulating through the ring until it returns to process P.

Process P then examines all process IDs present in the message.

The process with the highest priority number is selected as the new coordinator.

A coordinator message containing the selected process ID is then passed around the ring.

Each process receives the coordinator message and updates its coordinator information.

When the coordinator message completes one full cycle, all processes know the identity of the new coordinator.

If the coordinator fails again, the election process is repeated.

The ring algorithm ensures that only one coordinator is selected among all active processes.

It provides an organized way of leader election by passing messages in a circular manner.